

# 6th Grade Math Review

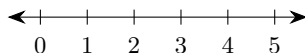
Name: \_\_\_\_\_ Date: \_\_\_\_\_

## Directions

**No calculator.** Write decimal answers as decimals. Write fraction answers in simplest terms. Show work in space provided or on scrap paper as directed.

### A. Number Lines

1. Plot the point 4 on the number line below:



### B. Order Of Operations

- |                                |                             |
|--------------------------------|-----------------------------|
| 1. $8 + 3 \times 4 =$ _____    | 4. $(5 + 1)^2 =$ _____      |
| 2. $(21 - 4) \times 3 =$ _____ | 5. $3(2^2) =$ _____         |
| 3. $2^4 + 1 =$ _____           | 6. $(3 \times 2)^2 =$ _____ |

### C. Decimals, Fractions, And Percents

- |                                               |                                       |
|-----------------------------------------------|---------------------------------------|
| 1. In 82.731, the tenths digit is _____       | 5. $65\% =$ _____ <i>as a decimal</i> |
| 2. $314.7 + 29.58 =$ _____ ( <i>decimal</i> ) | 6. $0.8 =$ _____ <i>as a fraction</i> |
| 3. $81.5 - 7.94 =$ _____ ( <i>decimal</i> )   | 7. $15\%$ of \$80 = _____             |
| 4. $0.32 =$ _____%                            |                                       |

### D. Facts And Decimal Multiply/Divide

- |                         |                                                |
|-------------------------|------------------------------------------------|
| 1. $9 \times 6 =$ _____ | 3. $2.4 \times 1.5 =$ _____ ( <i>decimal</i> ) |
| 2. $63 \div 9 =$ _____  | 4. $5.6 \div 0.8 =$ _____ ( <i>decimal</i> )   |

### E. Ratios, Fractions, And Proportions

- |                                           |                                                     |
|-------------------------------------------|-----------------------------------------------------|
| 1. Greater: $\frac{5}{8}$ or 0.61? _____  | 3. 4 snacks cost \$20. 7 snacks cost _____ dollars. |
| 2. $\frac{4}{7} = \frac{?}{21}$ ? = _____ | 4. $\frac{5}{6} = \frac{x}{18}$ $x =$ _____         |

### F. Vocabulary, Expressions, And Equations

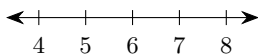
- |                                       |                                            |
|---------------------------------------|--------------------------------------------|
| 1. In $7m$ , 7 is the _____           | 4. Undo $\times 5$ , use inverse op: _____ |
| 2. In $n + 8 = 12$ , $n$ is the _____ | 5. Combine like terms: $6y - 2y =$ _____   |
| 3. Undo $+8$ , use inverse op: _____  | 6. If $n = 4$ , then $5n - 3 =$ _____      |

### G. Distributive Property

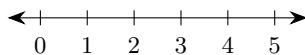
1. Expand:  $3(y + 6) =$  \_\_\_\_\_

### H. Equations, Inequalities, And Graphing

- |                                                       |                             |
|-------------------------------------------------------|-----------------------------|
| 1. $x + 7 = 18$ $x =$ _____                           | 3. Solve: $x + 3 > 8$ _____ |
| 2. $\frac{x}{4} = 6$ $x =$ _____                      | 4. Solve: $x - 4 < 2$ _____ |
| 5. Write an equivalent way to express $6 < y$ : _____ |                             |
| 6. Graph $6 < y$ on a number line:                    |                             |



7. Graph  $x > 2$  on a number line:



### I. Geometry

1. Rectangle with length 9 and width 4: Area = \_\_\_\_\_
2. Rectangle with length 11 and width 3: Perimeter = \_\_\_\_\_
3. Rectangle with area 42 sq units and width 6: Length = \_\_\_\_\_

## J. Units And Quantities

1. 3 hours = \_\_\_\_\_ minutes
2.  $\frac{3}{5}$  of 35 = \_\_\_\_\_
3. 210 miles/3 hours = \_\_\_\_\_
4. 72 inches = \_\_\_\_\_ yards
5. 150 miles/5 gallons = \_\_\_\_\_

## K. Expressions, Factors, And Fraction Structure

1. Is  $8x + 1$  a linear expression? \_\_\_\_\_
2. Prime factorization of 90 = \_\_\_\_\_
3. Find the GCF of 27 and 36: \_\_\_\_\_
4. Simplify  $\frac{27}{36} =$  \_\_\_\_\_
5. Convert  $2\frac{2}{3}$  to an improper fraction: \_\_\_\_\_
6. Which is greater:  $1\frac{3}{4}$  or  $(1)(\frac{3}{4})$ ? \_\_\_\_\_
7. In  $\frac{m}{2}$ , coeff of  $m$  is \_\_\_\_\_
8. In  $\frac{q}{5}$ , coeff of  $q$  is \_\_\_\_\_
9. Commutative:  $8 + 5 =$  \_\_\_\_\_
10. Commutative:  $7 \times a =$  \_\_\_\_\_
11. Associative:  $(b + 3) + 9 =$  \_\_\_\_\_

## L. Distributive Property And Factoring

1. Factor:  $8x + 16 =$  \_\_\_\_\_

## M. Decimal Computation

1.  $428.5 + 37.26 =$  \_\_\_\_\_ (*decimal*)
2.  $700.32 - 84.67 =$  \_\_\_\_\_ (*decimal*)
3.  $18.4 \times 12 =$  \_\_\_\_\_ (*decimal*)
4.  $168.8 \div 8 =$  \_\_\_\_\_ (*decimal*)
5.  $4.2 \times 1.5 =$  \_\_\_\_\_ (*decimal*)
6.  $7.2 \div 0.8 =$  \_\_\_\_\_ (*decimal*)

## N. Fraction Operations And Properties

1.  $\frac{1}{2} + \frac{1}{4} =$  \_\_\_\_\_ (*simplest*)
2.  $\frac{5}{6} - \frac{1}{3} =$  \_\_\_\_\_ (*simplest*)
3.  $\frac{5}{8} \times \frac{4}{15} =$  \_\_\_\_\_ (*simplest*)
4.  $\frac{7}{8} \div \frac{1}{4} =$  \_\_\_\_\_ (*simplest*)
5. Commutative:  $7 \times a =$  \_\_\_\_\_
6. Associative:  $(x + 5) + 8 =$  \_\_\_\_\_
7. Is subtraction associative? Compare  $(12 - 4) - 3$  and  $12 - (4 - 3)$ : \_\_\_\_\_